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SUMMARY

- Over 5 years of experience as a Data Analyst, specializing in machine learning, advanced data analytics, and statistical modeling to optimize investment strategies, enhance risk management, and drive business value.
- Expertise in developing and deploying predictive models using Python, R, and XGBoost, achieving significant improvements in risk prediction accuracy and portfolio performance.
- Skilled in conducting comprehensive risk analysis using advanced Excel functions, Analytical Solver, and time-series analysis techniques to identify and mitigate potential investment risks.
- Proficient in managing and processing large-scale datasets with Azure data services, ensuring data cleansing, structuring, and readiness for in-depth analysis.
- Designed and implemented ETL pipelines, transforming data from hierarchical to relational database systems to improve query efficiency.
- Developed robust data pipeline architectures within Azure and executed efficient ETL processes using MySQL, maintaining data integrity and accuracy.
- Created dynamic visualizations and dashboards in Tableau, transforming complex data insights into actionable recommendations for strategic decision-making.
- Created dynamic Tableau dashboards to visualize key metrics, enhancing data-driven decision-making and reducing operational costs.
- Utilized Bloomberg Terminal for comprehensive financial data analysis, trading capabilities, and market insights, supporting informed investment strategies.
- Developed predictive models for diabetes education staffing using Python, scikit-learn, and TensorFlow.
- Applied technical analysis tools such as RSI and MACD to identify trading signals, confirm trends, and detect potential market reversals.
- Executed day trading strategies with high liquidity stocks, leveraging real-time news updates and technical indicators to refine investment approaches.
- Extensive experience in data analytics, leveraging advanced SQL and Excel to query and summarize electronic medical records and regional data.
- Engineered machine learning algorithms, including regression models and neural networks, leveraging GPU acceleration for efficient training and optimization.
- Conducted stratified random sampling, one-hot encoding, and feature engineering using Python's Pandas, NumPy, and scikit-learn libraries for accurate model development.
- Employed cross-validation techniques and stratified K-fold validation to evaluate model performance, optimizing hyperparameters through exhaustive grid search.
- Defined and implemented conservative and aggressive credit approval strategies using advanced decision analysis techniques to balance risk and reward.
- Utilized JIRA and Confluence for efficient project management, issue tracking, and collaboration, enhancing the agile development process.
- Collaborated closely with business stakeholders and development teams to gather requirements, implement technical enhancements, and ensure project alignment.
- Provided data-driven insights and recommendations, leveraging historical records and application data to inform credit risk management strategies.
- Leveraged advanced SQL queries, Google Sheets, and Looker for KPI formulation and reporting, enhancing data-driven decision-making processes.
- Architected and optimized machine learning workflows in cloud environments using AWS SageMaker and Databricks, ensuring scalable and efficient model training, deployment, and monitoring.
- Integrated advanced natural language processing (NLP) techniques using NLTK and spaCy to analyze and extract insights from unstructured text data, enhancing sentiment analysis and topic modeling capabilities.
- Developed and maintained CI/CD pipelines for machine learning models using GitHub Actions and Jenkins, automating the deployment process and ensuring continuous integration and delivery of high-quality predictive analytics solutions.

- Executed cohort analysis and predictive analytics using R, Tableau, and Apache Spark, resulting in improved operational efficiency and increased revenue.

CERTIFICATIONS AND TECHNICAL SKILL EXPERIENCE	
Tools:	R Studio, Google Collaboratory, Eclipse, Azure, AWS Cloud, STATA.Hadoop Ecosystem (HDFS, YARN, Map Reduce), Apache Spark (with pyspark), Hive, HBase, Apache Airflow, Atlas, Tableau, Kubernetes, Power BI, MYSQL, MongoDB, Salesforce, Optimizely, MS Excel (Analytical Solver), Adobe Target, Google BigQuery, SAP.
Programming:	Python (NumPy, Pandas, Matplotlib, XGBoost, A/B Testing, Neural Network, Seaborn, K-means Clusterings), SQL, R, C, Java, HTML5, CSS3, PHP.
Software:	MS Office (Word, Excel, Outlook, PowerPoint), Jira, Spark, Jenkins.
Certifications:	Python, Java, DevOps Foundations, Microsoft Azure AI – 900, Data Science Foundations, Machine Learning, Cambridge English-Business English, Lean Six Sigma.

PROFESSIONAL EXPERIENCE

Data Analyst | Cardinal Health | USA

Apr 2023 – Present

Project Objective – To leverage advanced data analytics, machine learning algorithms, and data engineering techniques to enhance healthcare quality metrics, optimize operational efficiency, and support strategic initiatives through comprehensive analysis and dynamic visualization for the Cardinal Health Firm.

- Leveraged advanced SQL for querying electronic medical records and Excel for creating Pivot Tables to summarize regional data.
- Designed and implemented ETL pipelines to migrate and transform data from hierarchical to relational database management systems, increasing query efficiency by 30%.
- Utilized Python and machine learning algorithms for forecasting patient volumes and predictive modeling, leading to a 20% improvement in patient flow management.
- Created dynamic Tableau dashboards to visualize key metrics such as ALOS and patient census, facilitating data-driven decision-making and reducing operational costs by 15%.
- Applied data engineering techniques to ensure high data quality and integrity, supporting comprehensive analysis.
- Developed predictive models for diabetes education staffing and other healthcare quality measures using Python, scikit-learn, and TensorFlow.
- Collaborated with cross-functional teams to communicate insights and recommendations effectively, enhancing strategic planning.
- Conducted exploratory data analysis on complex datasets using Python, Pandas, and Jupyter Notebooks, providing actionable insights for operational efficiency.
- Consistently used advanced data analytics tools such as SQL, Tableau, and Python, along with machine learning algorithms like random forests and gradient boosting using libraries such as scikit-learn and TensorFlow, to improve healthcare quality metrics, operational efficiency, and strategic initiatives.

Business Data Analyst | Bank of America | NC, USA

May 2022 – Apr 2023

Project Objective – Developed and Deployed Machine Learning models for accurate credit risk prediction and optimized credit approval strategies, utilizing advanced data processing, feature engineering, and visualization techniques to support informed decision-making and align with business objectives.

- Conducted stratified random sampling using Python's Pandas and NumPy libraries to extract a representative 20% subset from two large-scale datasets.
- Applied one-hot encoding with scikit-learn to transform categorical variables and aggregated 13 months of historical data for feature engineering, incorporating time-series analysis techniques.
- Expert in designing and conducting A/B tests to optimize user experiences and marketing strategies using tools such as Google Optimize, Optimizely, and Adobe Target; proficient in statistical analysis techniques to ensure the validity and reliability of test results.
- Engineered a comprehensive feature set including lag features, rolling statistics, and interaction terms, leveraging XGBoost's feature importance metrics to retain predictors with an importance score above 0.5%.
- Developed and optimized XGBoost models using Python's xgboost library, employing exhaustive grid search for hyperparameter tuning.
- Experienced in setting up and managing comprehensive experimentation frameworks, including multi-armed bandit algorithms and Bayesian optimization methods, to continuously improve product features and marketing campaigns; utilized platforms like Split.io and LaunchDarkly for feature flagging and experiment management.

- Experienced in designing and implementing randomized controlled trials (RCTs) and quasi-experimental designs to assess the effectiveness of marketing strategies and interventions, ensuring robust and unbiased results.
- Customized QAD ERP modules to streamline operations and improve data flow across various business functions.
- Built and trained neural network models with Keras and TensorFlow, utilizing GPU acceleration for efficient training and model optimization.
- Led exhaustive grid searches for XGBoost and neural network models, recording evaluation metrics such as AUC, F1-score, precision, and recall in a performance-tracking table using Pandas DataFrame.
- Implemented cross-validation techniques and stratified K-fold validation to evaluate model performance, selecting the optimal model based on comprehensive performance metrics and business objectives.
- Defined and implemented conservative and aggressive credit approval strategies by setting tailored decision thresholds based on model output probabilities.
- Collaborated with the IT team to implement system upgrades and patches for QAD ERP, ensuring minimal disruption to business processes.
- Utilized advanced decision analysis techniques to balance risk and reward, aligning model predictions with strategic business objectives for credit approval processes.
- Employed JIRA for efficient ticket resolution and Confluence for creating detailed storyboards, enhancing collaboration and issue tracking.
- Proficient in applying causal inference techniques such as propensity score matching, synthetic control methods, and difference-in-differences analysis to isolate and measure the impact of interventions in observational data settings.
- Ensured compliance with industry standards and regulations through regular audits and updates within the QAD ERP system, maintaining operational integrity.
- Actively participated in Sprint Planning, daily scrum, and retrospective meetings, contributing to the agile development process and iterative project improvement.
- Developed dynamic Tableau dashboards to visualize credit card application data, defaults, and model predictions, supporting strategic decision-making.
- Collaborated closely with business stakeholders to gather requirements and with development teams to implement technical enhancements, ensuring project alignment and success.
- Provided data-driven insights and recommendations to the business team, leveraging historical records and application data to inform credit risk management strategies.
- Managed end-to-end model deployment, ensuring robust integration of predictive models into the credit risk analysis pipeline for real-time decision support.

Data Analyst | Emproto Technologies | India

Aug 2019 – Apr 2022

Project Objective - To leverage cutting-edge machine learning algorithms, advanced data analytics, and optimization models to significantly enhance operational efficiency, improve predictive accuracy, and support data-driven strategic decision-making within a complex business landscape.

- Spearheaded the development and deployment of machine learning models using Python, R, and Xgboost, resulting in a 15% improvement in risk prediction accuracy and significant annual cost savings.
- Leveraged Tableau and Excel to create dynamic dashboards, employing pivot tables and advanced VLOOKUP functions to transform raw data into actionable insights, which led to a 17% reduction in backorders.
- Utilized Analytical Solver for complex optimization and simulation models, enhancing risk assessment and mitigation strategies, resulting in a 20% improvement in operational efficiency.
- Successfully collaborated with product managers, engineers, and data scientists to implement A/B tests and experimentation protocols, ensuring alignment with business objectives and technical feasibility; leveraged Jira and Confluence for project management and documentation of experimentation processes.
- Skilled in leveraging data analytics to analyze brand marketing efforts, track key performance indicators, and provide actionable insights to enhance brand positioning and market penetration.
- Conducted comprehensive data mining and cleansing of structured and unstructured data using Pandas and NumPy, improving demand forecasting and reducing operational costs by 10%.
- Engineered and processed unstructured data using advanced clustering algorithms such as K-means, and hierarchical clustering, combined with regression models like Linear Regression. Employed scikit-learn and TensorFlow to transform raw data into structured formats, facilitating comprehensive analysis and deriving actionable insights for strategic decision-making.

- Solid understanding of various marketing channels (e.g., digital, social media, email, and traditional media) and metrics (e.g., conversion rates, customer acquisition cost, lifetime value) to evaluate and optimize marketing campaigns.
- Integrated neural network algorithms for high-stakes risk assessment projects, resulting in a 25% increase in predictive model performance and a significant reduction in financial risks.
- Designed and built statistical analysis models on large datasets using Apache Spark, leveraging Python and R for data processing and analysis. Implemented predictive analytics to increase online sales by up to 10% per product and reduce cart-abandonment rates.
- Partnered directly with executive teams to formulate and report on KPIs using advanced SQL queries, Google Sheets, and Looker, enhancing data-driven decision-making processes and strategic planning across the organization.
- Executed cohort analysis in collaboration with product managers using R and Tableau to uncover pricing adjustment opportunities, leading to an annual revenue increase. Leveraged SQL and Python for precise data extraction and analytical processing.
- Developed reusable ETL components using Pentaho Kettle and Apache NiFi, optimizing data pipeline efficiency, and reducing projected data analysis time by one week. Enhanced the overall efficiency of data processing workflows by incorporating best practices in data engineering and automation.

ACADEMIC PROJECT EXPERIENCE

IBM HR Analytics Employee Attrition and Performance | SQL

May 2022 – Jul 2022

- Designed and normalized a comprehensive database for the IBM HR Analytics Employee Attrition and Performance dataset from Kaggle, ensuring compliance with Third Normal Form (3NF) using an Entity-Relationship (ER) diagram.
- Created and managed the database in MongoDB, leveraging Mongo Compass and Atlas for efficient data import and administration.
- Transformed raw data into structured format, optimizing schema design for performance and scalability.

WHO COVID – 19 data analysis for Visualization based on region | Tableau

Jan 2023 – March 2023

- Utilized advanced geospatial segmentation in Tableau to analyze and visualize WHO COVID-19 data, delivering actionable insights for strategic decision-making.
- Integrated multiple datasets, performed complex data transformations, and leveraged advanced Tableau calculations and LOD (Level of Detail) expressions to enhance data granularity and visualization precision.

EDUCATION

The University of Texas at Dallas, Dallas, Texas

- Master of Science in Business Analytics – Data Science Track (Dean’s Excellence Scholar)

CGPA 3.78

R.M.K. Engineering College, Chennai, India

- Bachelor of Technology in Information Technology

CGPA 3.60